Antenatal Steroids and Tocolytics in Pregnancy

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KEYWORDS

• Tocolysis • Antenatal corticosteroids • Preterm birth • Preterm labor

KEY POINTS

- Antenatal corticosteroids reduce morbidity and mortality associated with preterm birth in neonates.
- Antenatal corticosteroids should be offered to all pregnant people at risk of preterm birth within 7 days who are between 24 weeks 0 days and 33 weeks 6 days gestational age.
- Antenatal corticosteroids can be offered to all pregnant people at risk of preterm birth within 7 days who are between 34 weeks 0 days and 36 weeks 6 days if they have not received a prior corticosteroid course.
- Additional data are needed to understand the long-term risks and benefits of antenatal corticosteroid administration in the periviable and late-preterm periods.
- Tocolysis can be considered to temporarily slow uterine activity and allow the administration of antenatal corticosteroids or transfer to a higher level of care for individuals at risk of preterm birth.

ANTENATAL CORTICOSTEROIDS

Introduction

Antenatal corticosteroid administration is one of the most important obstetric interventions to reduce the morbidity and mortality of the neonate associated with preterm birth and has become the standard of care for pregnant individuals with anticipated preterm delivery, particularly before 34 weeks gestational age.[1–4] Owing to the rising rates of preterm birth in the United States, with an estimated 12% to 15% of births occurring before 37 weeks' gestation, the administration of antenatal steroids has the potential to remarkably reduce morbidity in the neonatal period.[5,6]

As demonstrated in a large meta-analysis of 30 randomized controlled trials with over 8000 participants, infants whose mothers received antenatal corticosteroids have significantly reduced risks of developing respiratory distress syndrome (RDS) (relative risk [RR] = 0.66; 95% confidence interval [CI] 0.56–0.77), necrotizing

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enterocolitis (RR = 0.50; 95% CI, 0.32–0.78), intraventricular hemorrhage (RR = 0.55; 95% CI, 0.40–0.76), and neonatal death (RR = 0.69; 95% CI, 0.59–0.81) when compared with infants born at the same gestational ages to mothers who did not receive corticosteroids.[1–4,7,8]

Although the ultimate minimization of these morbidities likely lies in the reduction of preterm birth, both spontaneous and indicated, antenatal corticosteroid treatment remains a mainstay for the reduction of these morbidities.

Discussion

Physiology

After first demonstration in a sheep model in the 1960s, the benefits of antenatal corticosteroids have been established in dozens of human studies and have become the standard of care in the treatment of gravid people at risk of preterm delivery.[9] In the fetal lung, antenatal steroids primarily function by means of the induction of structural and biochemical changes to improve lung compliance, surfactant production, and gas exchange.[5,10] In the early preterm period, the effect is primarily through the induction of type II pneumocytes, which produce surfactant, a substance that reduces surface tension in the lung and prevents the collapse of alveoli during expiration.[5,6,10] In late preterm and early term infants, antenatal corticosteroids are also thought to have a significant effect on ENaC, sodium channels within the lung that facilitate clearance of fetal alveolar fluid.[5,6]

Choice of medication

The 2 steroids with demonstrated fetal benefit for prenatal use are dexamethasone and betamethasone. The 2 drugs are structurally similar epimers and differ in their biochemical structure only in the orientation of a single methyl group. In the United States, betamethasone is typically administered as a 1:1 mixture of 2 prodrugs: betamethasone acetate and betamethasone phosphate. Betamethasone phosphate is rapidly metabolized by dephosphorylation, while betamethasone acetate undergoes a slower deacetylation process. This allows for rapid effect of the betamethasone phosphate with a delayed metabolism and effect from betamethasone acetate. Because of this combination of rapid and delayed-onset actions, betamethasone is typically dosed at 24-hour intervals. Dexamethasone is usually administered as dexamethasone sodium phosphate, which in similar fashion to betamethasone phosphate is rapidly dephosphorylated and has a rapid onset of action. Because of this rapid metabolism, dexamethasone is usually dosed every 6 hours.[11]

Although there are some limited pharmacokinetic data, the optimal dosing of either medication is unknown, as the well-studied doses were chosen arbitrarily. Nonetheless, the standard dosing for betamethasone is 2 intramuscularly administered 12 mg doses, 24 hours apart. Dexamethasone is administered as 4 intramuscular doses of 6 mg, 12 hours apart. Neither has clearly been established as preferred over the other, with some studies favoring betamethasone and others favoring dexamethasone.[5,6]

Maternal effects

Both agents have demonstrated benefit in the reduction of complications of prematurity as previously discussed. From a maternal perspective, the most common side effect of either medication is hyperglycemia, which may require additional monitoring and blood glucose control. For people with pregestational diabetes, a continuous insulin infusion may be required for acceptable serum glucose control while receiving treatment with antenatal corticosteroids. Because of this transient hyperglycemia effect, screening for gestational diabetes should be delayed at least 1 week after administration of corticosteroids, if not completed before administration. A transient maternal leukocytosis has been associated with corticosteroid administration, but typically does not exceed $20,000 \times 10^3$ cells/mL.[6] A 2020 meta-analysis suggested that the rates of maternal death, chorioamnionitis, and endometritis are not affected by the administration of antenatal corticosteroids.[8]

Current recommendations

The American College of Obstetricians and Gynecologists (ACOG), the Society for Maternal-Fetal-Medicine (SMFM), and Royal College of Obstetricians and Gynecologists (RCOG) all support a course of antenatal corticosteroids under the following circumstances[1,12,13]:

- For pregnant people between 24 weeks 0 days and 33 weeks 6 days gestational age at risk of preterm delivery in the next 7 days
- For pregnant people between 24 weeks 0 days and 33 weeks 6 days gestational age at risk of preterm birth within 7 days who have received 1 prior course of antenatal corticosteroids and for whom at least 7 days have passed since the first course
- Consideration of a single course of antenatal corticosteroids for patients between 34 weeks 0 days gestational age and 36 weeks 6 days gestational age who have not received a prior course of corticosteroids, after discussion of the risks and benefits of late-preterm administration

Special Populations and Controversies

Repeat courses

As previously noted, a single rescue course of steroids is recommended for pregnant people who are less than 34 weeks gestational age, at risk for preterm birth within 7 days, and who have received 1 prior course at least 7 days prior. This is based on meta-analysis data demonstrating that there was no longer a statistically significant reduction in rates of RDS more than 7 days after exposure to antenatal steroids and that a repeat course reduces the risk of neonatal respiratory morbidity without significantly increasing the risk of other maternal and neonatal complications including neonatal death, neonatal major disability, maternal chorioamnionitis or sepsis, and cesarean delivery.[4,6,8,14,15]

Several studies have since examined serial courses of corticosteroids, at either 7- or 14-day intervals, and their effects on the neonate. Although scheduled serial courses likely reduce the risk of short-term respiratory morbidity, both 7- and 14-day serial courses have been associated with lower birth weight and smaller head size in infants exposed to multiple courses across several randomized-controlled trials.[5,16] In 2007, a Cochrane review, including more than 2000 participants, assessed repeat courses of betamethasone and found a reduction in occurrence of neonatal lung disease (RR 0.82, 95% CI 0.72–0.93) but did not demonstrate a difference in mean birthweight.[17] Long-term data are mixed; however, because of concerns regarding fetal growth and lack of demonstrated long-term benefit of scheduled repeat courses of corticosteroids, administering more than 2 courses during a single pregnancy is not currently recommended by any major organization in the United States.

Multiple gestation

Further study is needed to clearly demonstrate the benefits of antenatal corticosteroid administration in twin gestations; however, in the setting of likely benefit and unlikely harm, it is reasonable to administer antenatal corticosteroids to twin and higher order multiple gestations for the same indications as singletons.[1,4]

Administration in the periviable period

Corticosteroids in the periviable period, typically defined as 22 to 24 weeks gestational age, should be considered if treatment would be aligned with a family's goals of care for the neonate. Data from an NICHD Neonatal Research Network observational cohort demonstrated a reduction in neonatal death and neurodevelopmental impairment in infants exposed to corticosteroids who were born at 23 weeks through 25 weeks 6 days gestational age. No clear benefit was demonstrated in infants born 22 weeks 0 days to 22 weeks 6 days, and the outcomes in this group were poor regardless of exposure to antenatal steroids.[2] If resuscitation of an infant born in the periviable period is pursued, steroids should be offered, as they are likely to reduce complications in this population, while acknowledging that periviable birth is still associated with high rates of respiratory distress and long-term neurodevelopmental disability.

Administration in the late preterm period

As noted previously, a single course of corticosteroids in the late preterm period (34 0/7–36 6/7 weeks gestational age) should be considered for patients who have not received a prior course and who are at risk for delivery within 7 days. This recommendation was derived from the Antenatal Betamethasone for Women at Risk for Late Preterm Delivery (ALPS) trial, the largest trial at the time of publication assessing steroids in late preterm infants. This study randomized 2831 people at high risk for delivery between 34 0/7 to 36 6/7 weeks gestational age to receive a single course of antenatal corticosteroids or placebo. This group found a reduction in the primary composite outcome, which included any of: continuous positive airway pressure (CPAP) or high-flow nasal cannula for at least 2 hours, supplemental oxygen for at least 4 hours, mechanical ventilation, extracorporeal membrane oxygenation (ECMO), and neonatal death.[18] Exposed neonates also demonstrated lower rates of need for resuscitation at birth, transient tachypnea of the newborn (TTN), and bronchopulmonary dysplasia. No differences were seen in RDS, and exposed neonates were more likely to require treatment for hypoglycemia, although this was usually not severe. Meta-analyses including the various late preterm trials have reinforced these findings, and demonstrated reductions in RDS, mechanical ventilation, and TTN but an increase in neonatal hypoglycemia. [8,19]

Although exposure to antenatal corticosteroids in the early preterm period (before 34 weeks) has been associated with decreased risk of long-term neurodevelopmental impairment, fewer data are available regarding exposure in the late-preterm period. Animal studies have suggested that exposure to steroids in the late preterm period may alter fetal brain development, with structural effects seen primarily in the hippocampus in addition to altered postnatal behaviors.[20] In people, a retrospective cohort study of Finnish children analyzed term born sibling pairs and found that children exposed to antenatal steroids were more likely to exhibit any childhood mental or behavioral disorder compared with their unexposed siblings, although gestational age at administration of steroids was not known.[21] Long-term follow-up data from randomized trials of a single course of late preterm steroids are scarce, but the available data are reassuring.

In 2013, a long-term follow-up study of the Antenatal Steroids for Term Elective Caesarean Section (ASTEC) trial demonstrated no differences in behavioral, cognitive, or developmental outcomes among school-aged children whose mothers received a single course of antenatal corticosteroids before delivery at 37 to 38 weeks' gestation.[22] The 30-year follow-up of the first landmark trial demonstrating effectiveness of betamethasone for reduction of neonatal morbidity examined 192 adults (87 exposed to antenatal betamethasone, 105 exposed to placebo). No differences were found on neurocognitive assessment between groups exposed or unexposed to betamethasone, including cognitive function, working memory and attention, psychiatric morbidity, handedness, and health-related quality of life. Notably, most of these participants were exposed in the late-preterm period during fetal life.[23,24] The ALPS neurocognitive follow-up study is currently being conducted, and results are expected by 2023.

Summary

Antenatal corticosteroids are highly effective to reduce neonatal morbidity and mortality associated with preterm birth.

Clinics Care Points

- Antenatal corticosteroids can reduce morbidity and mortality of preterm birth in neonates, including RDS, necrotizing enterocolitis, intraventricular hemorrhage, and neonatal death.
- Antenatal corticosteroids cause transient maternal leukocytosis and hyperglycemia but do not appear to increase maternal morbidity, including chorioamnionitis, sepsis, endometritis, and maternal death.
- Antenatal corticosteroids should be offered to all pregnant people at risk of preterm birth within 7 days who are between 24 weeks 0 days and 33 weeks 6 days gestational age.
- Antenatal corticosteroids can be offered to select pregnant individuals at risk of preterm birth within 7 days who are between 34 weeks 0 days and 36 weeks 6 days if they have not received a prior corticosteroid course.
- Additional data are needed to understand the long-term risks and benefits of antenatal corticosteroid administration in the periviable and late-preterm periods, particularly regarding long-term neurodevelopment in the late-preterm exposed fetuses.

TOCOLYTICS

Introduction

Preterm labor, diagnosed in the setting of regular uterine contractions with cervical change before 37 weeks gestation, accounts for approximately 50% of preterm births in the United States. As such, much research has gone into identifying predictors, methods of prevention, and possible treatments for preterm labor in order to prevent the known neonatal mortality and morbidity associated with it.[25] Tocolytics, medications administered to inhibit uterine contractions, have been studied for short- and long-term use for the prolongation of pregnancies affected by preterm labor.

Overall, less than half of women who are hospitalized for preterm labor ultimately deliver preterm, so identifying which women will benefit from short-term tocolysis to allow for the administration of antenatal corticosteroids is challenging. As a general rule, tocolysis should be used when the fetal risks of prematurity outweigh the fetal and maternal risks of prolonging a pregnancy in the setting of preterm labor. Typically, tocolysis is only used prior to 34 weeks 0 days gestational age.[25]

Discussion

Clinical use

A number of different agents have been studied to inhibit uterine contractions and those with the most data include calcium channel blockers, nonsteroidal anti-inflammatory drugs (NSAIDs), beta-adrenergic receptor agonists, and magnesium sulfate.[25]

Overall, the evidence supports the short-term use of tocolytic agents to allow the administration of antenatal corticosteroids, magnesium sulfate for fetal neuroprotection, and possible transfer to a higher acuity care setting for maternal and neonatal benefit.[25,26] Long-term use of tocolytics (ie, over more than 48 hours or for consecutive weeks) has not been proven to be effective and is generally avoided because of risks of maternal and fetal adverse effects.[27]

Tocolysis is not recommended beyond 34 weeks 0 days gestational age and is typically not recommended prior to 24 weeks 0 days gestational age unless neonatal resuscitation is being considered before that time.[26] Additional contraindications to tocolysis include intrauterine fetal demise or lethal fetal anomaly, heavy vaginal bleeding or suspected placental abruption, nonreassuring fetal status, severe preeclampsia or eclampsia, maternal instability, and chorioamnionitis.[25]

Calcium channel blockers

Calcium Channel blockers, most commonly nifedipine, inhibit voltage-gated calcium channels in smooth muscle cells, resulting in decreased release of stored calcium, which leads to myometrial relaxation via inhibition of calcium-dependent myosin light-chain kinase phosphorylation.[6]

A 2003 Cochrane review supported the use of calcium channel blockers as short-term tocolytics over other available agents (primarily beta-mimetics), with significantly reduced rates of preterm birth occurring within 7 days of treatment (RR = 0.76; 95% CI, 0.60–0.97) and before 34 weeks (RR = 0.83; 95% CI, 0.69–0.99).[28] More recently, a small randomized, placebo-controlled trial of 90 participants found no significant difference in the rate of preterm birth before 37 weeks (52% vs 48%, RR 1.1, 95% CI 0.7–1.7), or in delivery at least 48 hours from randomization (78% vs 71%, RR 1.1, 95% CI 0.9–1.4) for people who presented in preterm labor and were randomized to receive either nifedipine or placebo for 48 hours, although this study was likely underpowered to detect small differences in rates of preterm birth.[29]

Overall, although there are some conflicting data, the sum of the evidence supports the short-term use of calcium channel blockers for tocolysis before 34 weeks' gestation.[25]

Nonsteroidal anti-inflammatory medications

Nonsteroidal anti-inflammatory drugs (NSAIDs) are used to reduce prostaglandin synthesis via inhibition of cyclooxygenase (COX). Prostaglandins are known to mediate uterine muscle contraction and are used for ripening the cervix and induction of labor.[6]

A 2005 Cochrane Review concluded the COX inhibitors (primarily indomethacin) were associated with a reduction in preterm birth before 37 weeks gestation, but that data were insufficient to recommend use because of overall small numbers and the potential for adverse fetal effects.[30] Reported adverse effects associated with NSAID use in pregnancy include maternal nausea, gastritis, and renal impairment, as well as fetal risks including oligohydramnios, premature closure of the ductus arteriosus, and increased risk of necrotizing enterocolitis in preterm neonates. These risks can be mitigated by restricting indomethacin use to pregnancies before 32 weeks gestation and limiting use to 48 hours of therapy.[6,25]

Magnesium sulfate

Magnesium sulfate acts at the motor end plate and the cell membrane by competition with calcium and resulting inhibition of myometrial contractility.[6]

Magnesium sulfate, despite having limited evidence of benefit, is often used as a tocolytic owing its benefits for fetal neuroprotection.[6,31,32] A 2002 Cochrane meta-analysis including over 2000 women found no difference in rates of preterm birth within 48 hours (RR 0.85, 95% CI 0.58–1.25) or before 34 or 37 weeks in women given magnesium sulfate over controls, and the rates of neonatal death were significantly increased in those who received magnesium therapy (RR 2.82, 95% CI 1.20–6.62).[33] This association with neonatal death is poorly understood. Two additional randomized trials comparing magnesium sulfate to nifedipine and celecoxib (COX inhibitor) found no difference in the number of women who delivered within 48 hours.[6]

Because of potentially serious maternal adverse effects including the development of pulmonary edema or hypotension, the use of concurrent nifedipine or terbutaline tocolysis in combination with magnesium sulfate should be cautioned.[25]

Beta-adrenergic receptor agonists

Terbutaline, a beta agonist occasionally used for tocolysis in the United States, acts via beta receptors in the myometrium to relax smooth muscle and inhibit myometrial contractions.[6] Owing to the widespread presence of beta receptors throughout the body, terbutaline has many reported side effects (**Table 1**) and thus is not recommended for long-term and repeated use.[6,25]

In 2011, the US Food and Drug Administration (FDA) issued a warning regarding the use of terbutaline for the treatment of preterm labor because of serious maternal adverse effects including maternal cardiac events and death.[34] Neither ACOG nor the most recent Cochrane review support the use of terbutaline for longer than 48 hours, and this medication should generally be reserved for the treatment of acute uterine tachysystole.[6,35]

Maintenance therapy

The only tocolytic medication that has demonstrated efficacy in the prolongation of pregnancy as a maintenance therapy is atosiban, an oxytocin and vasopressin receptor inhibitor. This medication is not available in the United States.[36] As mentioned previously, because of the risk of maternal side effects and lack of evidence supporting efficacy, tocolysis is typically reserved for short-term use in the United States, although atosiban is still frequently used in other parts of the world.

SUMMARY

Several agents are used for short-term tocolysis, primarily with the goal of obtaining sufficient time for administration of antenatal corticosteroids or transfer of care to a higher acuity care center. The data in support of maintenance tocolysis are poor, and data supporting short-term use are mixed with regard to efficacy.

CLINICS CARE POINTS

- Short-term tocolysis with calcium channel blockers and NSAIDs (before 32 weeks' gestation) can be used for the short-term prolongation of pregnancies in the setting of preterm labor.
- Magnesium sulfate, although beneficial for fetal neuroprotection, should be avoided for tocolytic therapy alone, and combined use with calcium channel blockers and beta mimetic agents is cautioned because of the risk of maternal side effects.
- Maintenance tocolysis for greater than 48 hours is not recommended.

Table 1
Tocolytic medications and dosages in pregnancy

Agent or Medication Class	Typical Dosing	Fetal Adverse Effects	Maternal Adverse Effects	Contraindications
Calcium Channel Blockers	Nifedipine: 10–20 mg every 3–6 h	None known	Hypotension, headache, nausea, dizziness	• Hypotension • Severe maternal cardiac disease
NSAID	Indomethacin: 50 mg load (rectally or orally) followed by 25 mg every 6 h for up to 48 h	• Oligohydramnios • Increased risk of necrotizing enterocolitis in preterm neonates • Premature closure of the ductus arteriosus	Nausea, kidney injury, gastritis	• Gestational age ≥ 32 weeks • Maternal bleeding disorder or thrombocytopenia • Gastrointestinal ulcer • Caution with maternal renal disease and maternal history of bariatric surgery
Oxytocin receptor antagonist	Atosiban: 6.75 mg bolus injection followed by 300 μg/min infusion for 3 h, followed by 100 μg/min infusion for up to 45 additional hours	None known	Injection site reactions	• No absolute contraindications • Not available in the United States
Beta-adrenergic receptor agonists	Terbutaline: 0.25 mg subcutaneous injection every 4 h	Neonatal hypoglycemia, fetal tachycardia	Tachycardia, palpitations, tremor, headache, hypokalemia, hyperglycemia	Maternal cardiac disease with sensitivity to tachycardia
Magnesium sulfate	Intravenous $MgSO_4$: 4 or 6 g bolus followed by 1–2 g/h for up to 12 h	None known, possible temporary neonatal depression although data are inconsistent	Flushing, headache, respiratory depression, cardiac arrhythmia or arrest, pulmonary edema	Myasthenia gravis *Caution with renal insufficiency because of impaired clearance

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